Hw4

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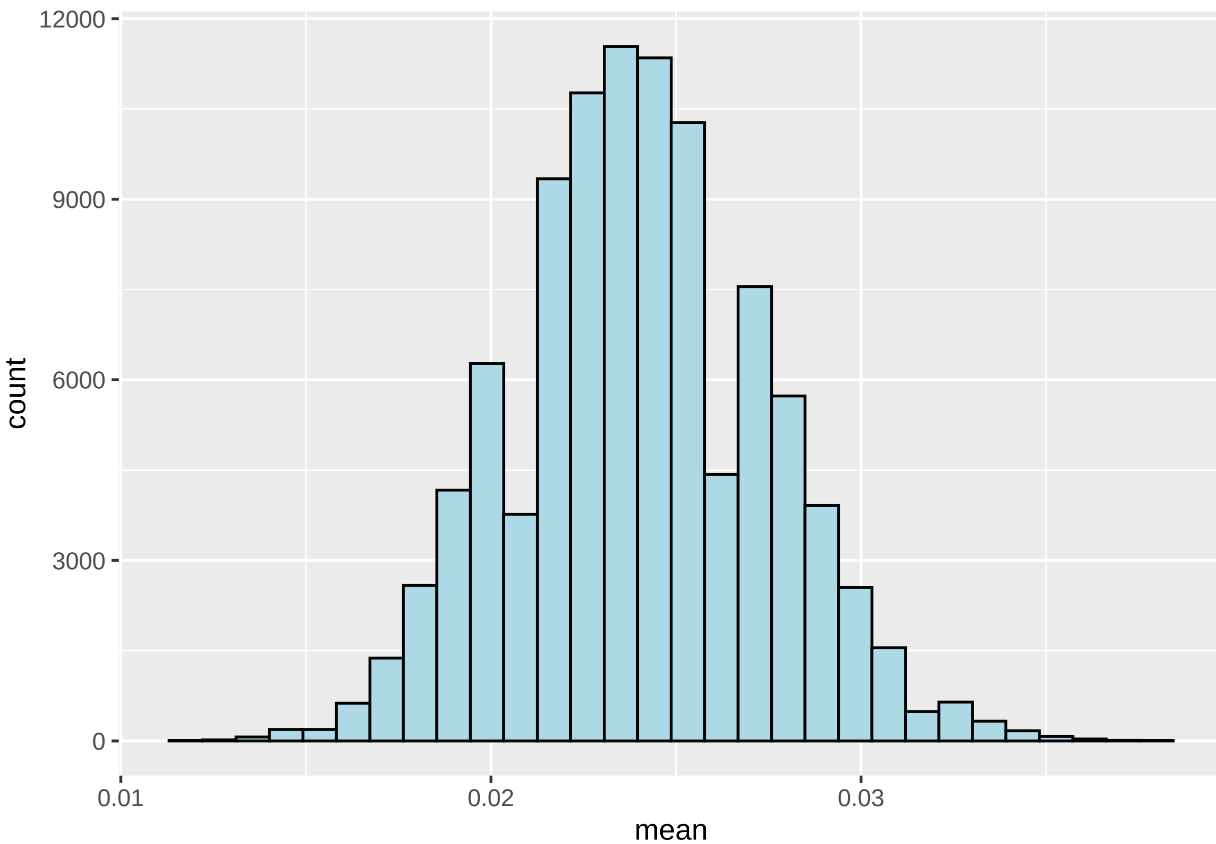
Link to github

Question 1

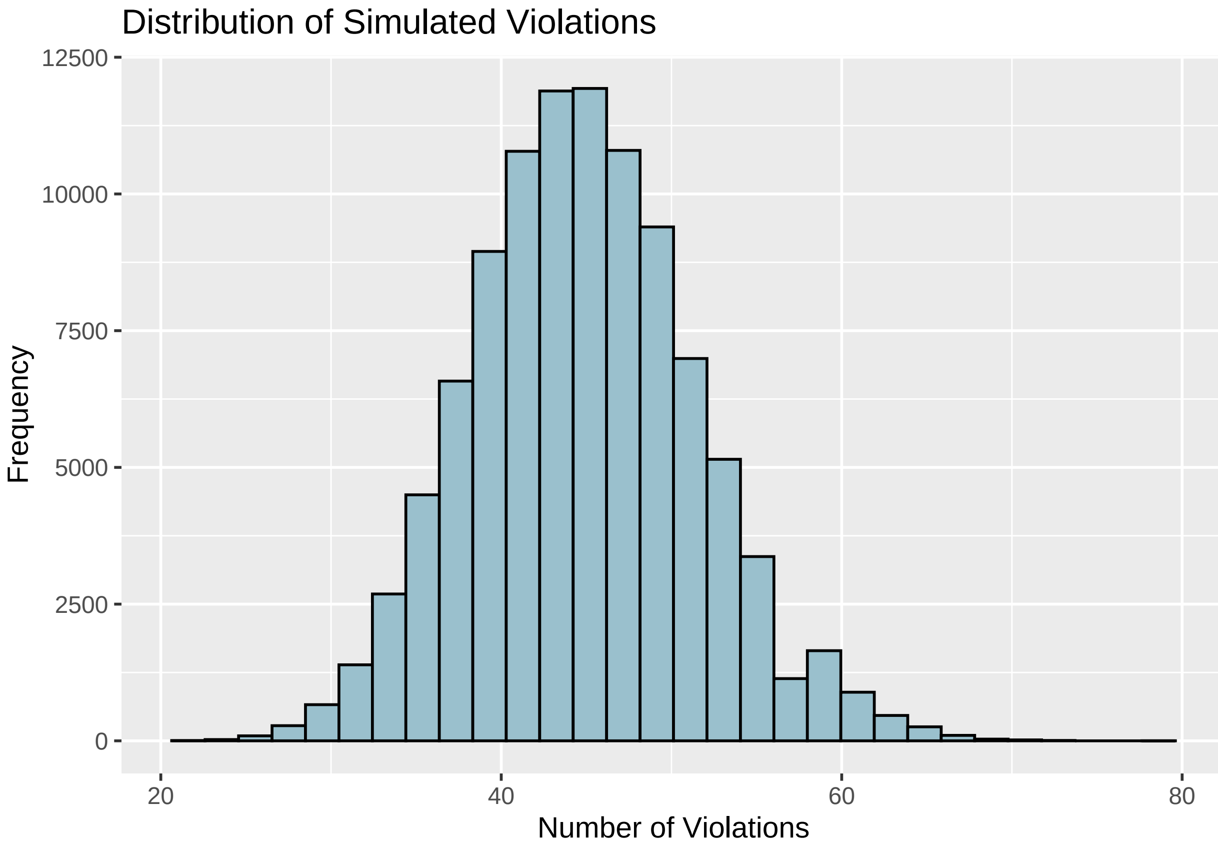



```
## [1] 0.002
```

The null hypothesis being tested is that if there is a probability of 2.4% trades being flagged, then there is something illegal going on and there should be an investigation into the company.

To measure against the null hypothesis, there will be the monte-carlo simulation with a re sampling that calculates the mean of each sample. This would mean that there would be a proportion of how many iterations exceed the desired ratio and that can be filtered out using the p value constraint. As the p value is around `0.002`, this is a fairly small threshold and therefore can say that the null hypothesis is rejected. This would support the alternative hypothesis that their trading activity is unusual and indicates a possibility of insider trading.

Question 2




```
## [1] 0.245
```

In this case, the null hypothesis is to check if there is a standard 3% citation baseline for being flagged due to random chance.

To measure against the null hypothesis, there will be the monte-carlo simulation with a re sampling that calculates the mean of each sample. This would mean that there would be a proportion of how many iterations exceed the desired ratio and that can be filtered out using the p value constraint.

The P value differs greatly in this case as it measure to be `0.244`. This would mean that there is a 24.4% chance that there can be 8 or more violations due to chance. Since this probability is relatively high, there is not enough evidence to conclude that there's a high violation rate. In this case, the null hypothesis is accepted.

Question 3

```
## [1] 0.014
```

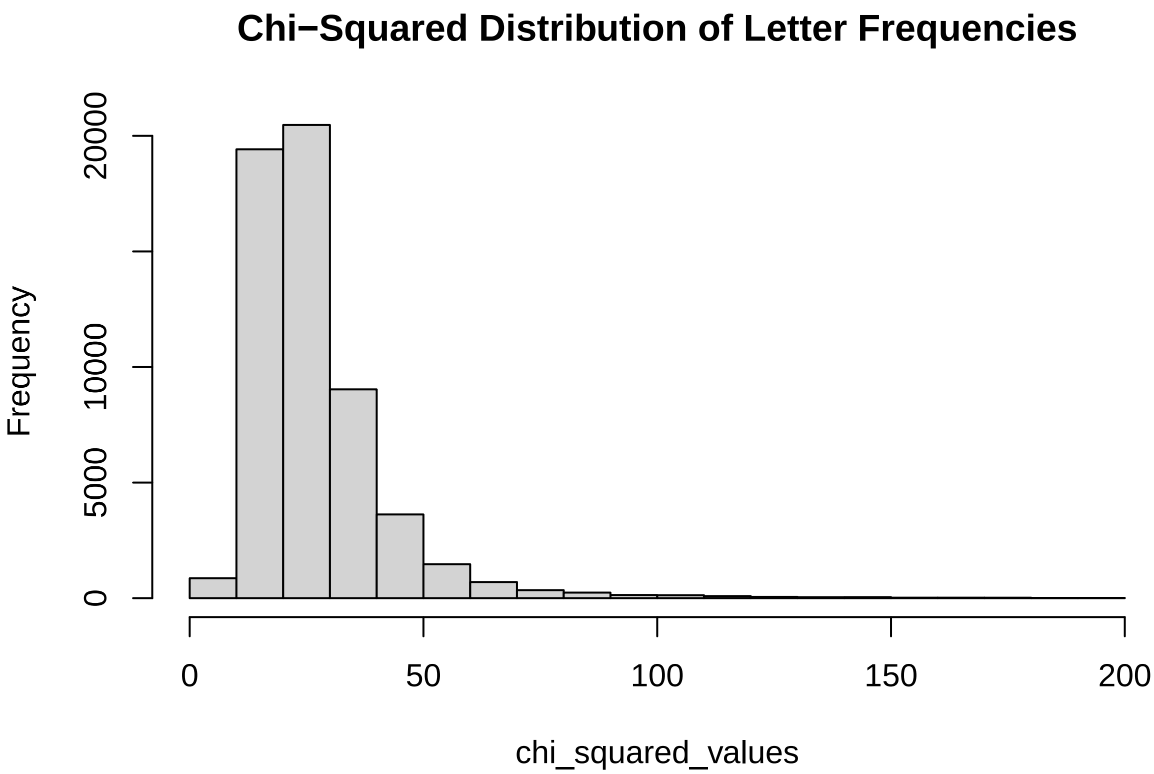
The null hypothesis in this case would be that the judge's selection process does not systematically change the representation of its group.

Since there is a multinomial setting, it would be best to use a chi-squared test to determine statistical significance. Then using the chisq function, there can be a detailed p value that is used to back up the claim.

After running the code, the chi-squared result was 12.426. This would mean that there is quite the difference between the observed and expected values. This difference would suggest there should be a rejection of the null. Furthermore, this is supported by the low p value of `0.014`. In conclusion, the judge's selection process does systematically change the representation of its group.

Question 4

Part A




```
## [1] 26.95
```

Part A creates a null hypothesis to test the LLM watermark on. It seems like the distribution is centered around a mean of around 27. This information can be useful in Part B.

Part B

```
##    Index P_Value
## 1      1     0.8
## 2      2     0.9
## 3      3     0.3
## 4      4     0.7
## 5      5     0.6
## 6      6     0.1
## 7      7     0.5
## 8      8     1.0
## 9      9     0.4
## 10    10     0.2
```

After taking the Chi-Squared difference between the expected (null) and the outcome (10 sentences), there is a clear outlier in which the p value is less than 0.05 which is sentence 6. This sentence is the following: `Feeling vexed after an arduous and zany day at work, she hoped for a peaceful and quiet evening at home, cozying up after a quick dinner with some TV, or maybe a book on her upcoming visit to Auckland.` As it has a p-value lower than 0.05, it would mean that there is a very low chance the null hypothesis occurs, therefore hinting at the fact that it is generated by an LLM.